

Compiling Local Update Contracts into Controller-Preserving Policies

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Updating a controlled system can require components to change at different times while old and new requirements overlap or exclude one another. Choosing an update sequence is insufficient when its steps have multiple possible outcomes. We present Fine-Grained Dynamic Update Controller Synthesis (FG-DUCS), which compiles local state transfers, individual requirement lifetimes and monitor initialization into a policy between supplied verified controllers. The policy must safely finish from every admitted starting state and for every possible outcome. We prove that the generated game implements the local contract and preserves the chosen new controller's behavior after handover. A sound and complete lazy procedure returns a policy with a progress certificate or proves that none exists. Complete examples show how merging components or requirement boundaries can prevent completion; a reverse example shows that merging can also help. An MTSA implementation decides 26/27 adapted contracts within 1,200 s, compared with 17 for eager successors and 14 for full construction. Five constructed families exhibit granularity-dependent feasibility; inherited inputs show no feasibility change. Commands and handover require uncontrollable quiescence, and the contract guarantees finite completion without fairness, rather than service continuity or elapsed-time bounds.

CCS Concepts: • **Software and its engineering** → **Software evolution**.

Additional Key Words and Phrases: dynamic software updating, discrete-event controller synthesis, supervisory control, on-the-fly synthesis

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1 Introduction

Verified endpoints leave their intervening update unspecified [3, 6, 13, 33]. Consider replacing two stations while retaining exactly one workpiece. Each replacement needs its station empty. Replacing both together is impossible, but moving the workpiece between separate replacements can make the update possible. A second choice concerns requirements: audit coverage may require the new audit rule before the old rule ends, whereas mutually exclusive access roles require the old role to end first. One global start/stop order cannot satisfy both. These are different reasons to expose update granularity.

An update may begin in any reachable old state, with environment-chosen events and transfer outcomes. Initial memory also matters: starting a monitor must not clear an existing inspection duty. A safe policy may inspect forever. The required result must cover every admitted entry and outcome, preserve declared obligations and force finite completion.

Our result compiles local update contracts into controller-preserving policies. Components, transfers, requirement lifetimes and initializers generate the mixed-state game between verified endpoints. Neither that graph nor an update sequence is supplied. Requirement intent and transfer code remain inputs; modeling effort is unmeasured.

DUCS [29] generates update and subsequent control from global transfers and transition requirements [30, Defs. 4.1–4.3]; GR(1) updating constructs its new controller and bridge independently and runs the new controller afterward [2, Def. 3.1, Sec. 4]. Reconnection to subsequent control is

therefore already established. FG-DUCS adds local transfers, independent boundaries and initializers to supplied endpoints and targets. Sections 3.4 and 7 compare generation responsibilities and OTF implementations.

C1: Contract compilation. A local input semantics with individual requirement boundaries, its generated game, and proofs of correspondence and endpoint-continuation preservation, with an initialization criterion for pure invariants (Sections 3 and 5.1).

C2: Certifying synthesis. A sound and complete specialized lazy procedure with checkable WIN/LOSS certificates (Section 4).

C3: Granularity boundaries. Complete component and requirement examples, sufficient merging obstructions and a reverse separation; Rolling supplies an exact batch bound (Sections 2 and 6.2).

C4: Implementation evidence. An MTSA tool evaluated on 27 adapted contracts and 55 constructed comparisons, retaining all adverse outcomes (Section 6).

2 From a Complete Contract to a Completing Policy

Component granularity chooses what is replaced together; requirement granularity chooses which guarantees start or end together. Two contracts separate these choices.

2.1 Cell: Complete Local Input and Whole Policy

Consider a constructed two-station cell with exactly one workpiece [20, 30]. Both transfers require an empty station. A keeps its behavior; new B requires inspection before departure, while the old rule forbids inspection. Ordinary events move or inspect the workpiece. New-start before old-stop prevents an unprotected gap; a pending inspection waits for old-stop. Inspection can repeat safely forever.

Table 1 specifies every component, endpoint and monitor. The interval requires exactly one workpiece, which plant transitions preserve. All labels are controllable; new-start before old-stop is the only dependency. Transfers are $g_A = g_B = \{(e, e)\}$, where e means empty.

Table 1. Complete Cell components, controllers and monitors. o/n denote old/new versions; h/e mean holding/empty. Parentheses mark initial states. m_A, m_B move to A, B ; j inspects at B .

Object	States (initial)	Ordinary transitions
A^o, A^n	$h, e (h)$	$h \xrightarrow{m_B} e, e \xrightarrow{m_A} h$
B^o	$h, e (e)$	$e \xrightarrow{m_B} h, h \xrightarrow{m_A} e$
B^n	$h, e (e)$	$e \xrightarrow{m_B} h, h \xrightarrow{m_A} e, h \xrightarrow{j} h$
C^o	$c_o (c_o)$	$c_o \xrightarrow{m_A, m_B} c_o$
C^n	$c_0, c_1 (c_0)$	$c_0 \xrightarrow{m_A, j} c_0, c_0 \xrightarrow{m_B} c_1, c_1 \xrightarrow{j} c_0, c_1 \xrightarrow{m_B} c_1$
t_{one}	$A, B, \text{err} (A)$	$A \xrightarrow{m_B} B, B \xrightarrow{m_A} A, A \xrightarrow{m_A} \text{err}, B \xrightarrow{m_B} \text{err}$
t_{old}	$k, \text{err} (k)$	$k \xrightarrow{j} \text{err}$
t_{inspect}	$c, d, \text{err} (c)$	$c \xrightarrow{m_B} d, d \xrightarrow{j} c, d \xrightarrow{m_A} \text{err}$

k, c, d mean old-safe, inspection-clear and inspection-pending; c_0, c_1 are distinct controller states. Component/controller alphabets are exactly their displayed labels; unlisted edges are absent. Unlisted non-error monitor edges self-loop, and err absorbs every label. Each comma-separated label denotes an edge.

Endpoint tuples list A, B , controller and its requirement monitor. The old initial tuple is $x_A = (h, e, c_o, k)$ and its other reachable tuple is $x_B = (e, h, c_o, k)$; both are entries. The new initial tuple

is $z_A = (h, e, c_0, c)$; its other reachable tuples are $y_B = (e, h, c_1, d)$ and $z_B = (e, h, c_0, c)$. Only $\{z_A, z_B\}$ are loadable targets: inspection must be clear at handover. The interval initializer records the holder in either all-old tuple. The new initializer admits all eight version-tagged one-workpiece tuples, installing d if B holds and c otherwise. These data specify the contract without an update sequence.

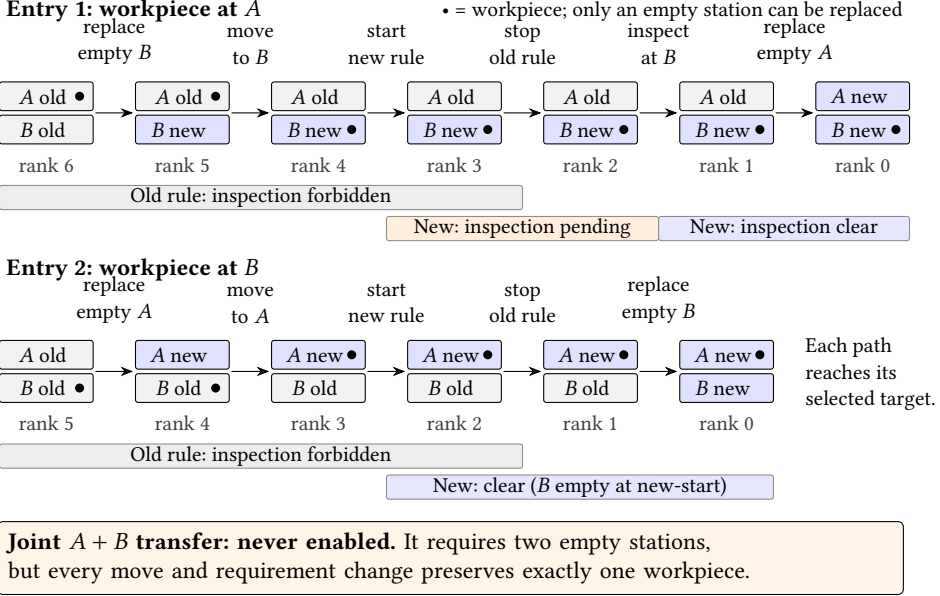


Fig. 1. A complete local policy from both old entries. Bands mark active rules and new-rule inspection memory; new-start precedes old-stop. Each arrow has one action and one outcome, decreasing rank toward its selected new target. Joint transfer merges these local commands with endpoints and requirements unchanged; this is not a DUCS encoding.

Feasibility and completion. Moving the workpiece enables local replacements; joint replacement requires both stations empty, impossible under any move or boundary. Endpoints and requirements remain fixed. Inspection may repeat forever safely, but every policy arrow decreases rank from six or five to zero, proving all-outcome handover from both entries.

A whole policy from the local input. Figure 1 covers both entries and targets: 13 states and 11 edges. Shown events have singleton outcomes; other controllables are disabled. Activation on the first path creates an inspection duty cleared before handover to z_B ; the second reaches z_A . Technical Appendix B and S1 give paths and closure.

2.2 Policy: Scheduling Requirement Lifetimes

Policy(2) separates requirement boundaries. Four bits record active old/new audit rules a_o, a_n and roles r_o, r_n . The interval invariant $(a_o \vee a_n) \wedge \neg(r_o \wedge r_n)$ forces new-audit start before old-audit stop, but old-role stop before new-role start. No order is supplied. The plant has one state in each version and all ordinary self-loops; transfer is identity. Each endpoint alternates its version-specific log and service, with its own access event enabled in either state. Its audit monitor rejects service without a log, and its role monitor rejects the other version's access. New audit starts unlogged

and new role starts safe. The old endpoint supplies logged and unlogged entries; one service event first clears a logged entry. A whole policy then executes

$$\text{start}_A, \text{stop}_A, \text{stop}_R, \rho, \text{start}_R,$$

where A denotes audit, R role and ρ an identity component transfer. Ranks six and five decrease to the compatible new endpoint; other controllables are disabled.

If both starts are merged and both stops are merged, the first start overlaps roles and the first stop removes audit coverage. Ordinary events and transfer cannot change these active bits, so neither entry has a completing merged policy. Technical Appendix I supplies every plant, controller, monitor, initializer and target, plus the whole policy and losing closure. This isolates boundary granularity; it does not assert greater expressiveness than DUCS or service continuity.

3 Defining an Update Contract

The input supplies event control, fixed endpoints, components/transfers, old/new/interval requirements, initializers, optional precedence and handover targets, without prescribing a sequence. Section 5.1 connects initializers to requirement scopes.

The choices and their costs. Admission releases old scheduling and liveness, retaining each old safety monitor until its stop. Local commands require *global* UC quiescence; perpetual UC activity can prevent completion. Service may be withheld, so uninterrupted service and bounded downtime are not promised. Fixing the new controller preserves selected continuations but restricts completion states.

3.1 Supplied Choices

Finite LTSs and safety monitors. A labeled transition system (LTS) $\mathcal{L} = (S, \Sigma_{\mathcal{L}}, \delta, s^0)$ has nondeterministic transitions $\delta : S \times \Sigma_{\mathcal{L}} \rightarrow 2^S$. Partition events as $\Sigma = \Sigma_{uc} \uplus \Sigma_c$: controllers may disable controllables but cannot select outcomes or suppress uncontrollable (UC) events. A synchronous-product event requires a transition in every component whose alphabet contains it; other components retain their state [22]. A monitor is an automaton that checks a requirement as events occur. Requirement r has a total deterministic monitor $t_r = (M_r, \Sigma, \eta_r, m_r^0, F_r)$ with absorbing errors for violating prefixes [1, 19]. The empty monitor product has one non-error state. Monitors observe labels; policies also observe successor component states.

Components and transfers. For each $i \in I$, let $e_i^{\text{old}}, e_i^{\text{new}}$ be the pre-update and post-update components and let $g_i \subseteq S_{e_i^{\text{old}}} \times S_{e_i^{\text{new}}}$ be their state-transfer relation. Define $g_i(s) = \{s' \mid (s, s') \in g_i\}$. The supplied g_i includes every modeled transformation outcome. Let $E^v = \parallel_{i \in I} e_i^v$.

Requirements and activation. Let pairwise-disjoint $R^{\text{old}}, R^{\text{new}}, R^{\text{upd}}$ contain the pre-update, post-update, and update-interval safety requirements, respectively. For each $r \in R^{\text{new}} \cup R^{\text{upd}}$, the input supplies a partial *monitor-initialization map* ι_r from version-tagged component tuples to $M_r \setminus F_r$. Its domain specifies where activation is allowed; in the cell, initialization records whether inspection is pending. For history-based requirements, residual soundness (RS) requires every continuation accepted by the installed state to be safe after every declared activation history. Activation-scoped invariants instead retain current facts and constrain behavior from their start; both interpretations are defined in Section 5.1. For $r \in R^{\text{upd}}$, $\text{dom } \iota_r$ includes every reachable old component state.

Endpoints and handover targets. For $v \in \{\text{old}, \text{new}\}$, the supplied finite controller C^v preserves uncontrollables and makes $E^v \parallel C^v$ deadlock-free and satisfy R^v . Hiding controller auxiliary events leaves observation alphabet $\bigcup_i \Sigma_{e_i^v}$; the union over both versions is the ordinary alphabet Σ_N . Write

Reach for initial-state reachability and

$$CL^v = E^v \parallel C^v \parallel \parallel_{r \in R^v} t_r.$$

Monitors are passive: every endpoint transition originates in $E^v \parallel C^v$; a monitor only updates on that label and adds no enabled event. The input supplies nonempty, checked targets $Z_{\text{load}} \subseteq \text{Reach}(CL^{\text{new}})$ with component, controller and new-monitor states. FG-DUCS keeps the supplied new controller fixed. The nine adapted benchmark inputs admit all of $\text{Reach}(CL^{\text{new}})$; constructed contracts may select a proper subset, as Cell does. For a supplied maximally permissive controller this is its reachable winning region, equal to the full region only if all winning states are represented and reachable.

Update commands and order. The set of update events, each executed exactly once, is

$$\Sigma_{\text{upd}} = \{\rho_i \mid i \in I\} \cup \{\text{stop}(r) \mid r \in R^{\text{old}}\} \cup \{\text{start}(r) \mid r \in R^{\text{new}}\}. \quad (1)$$

These controllable events transfer components, stop old monitors or start new ones, respectively, and are disjoint from ordinary events Σ_N : $\Sigma = \Sigma_N \uplus \Sigma_{\text{upd}}$ and $\Sigma_{\text{upd}} \subseteq \Sigma_c$. Each transfer event ρ_i is one atomic model transition: the controller chooses the update, but cannot choose its g_i outcome. They require uncontrollable quiescence: no modeled uncontrollable event may be enabled when an update command is issued.

An optional strict partial order $< \subseteq \Sigma_{\text{upd}} \times \Sigma_{\text{upd}}$ expresses precedence among update operations. For a pending-event set P , event a is order-enabled exactly when $a \in P$ and every $a' < a$ has left P . Monitors self-loop on update events that they do not specify.

Eligible contracts. Eligibility requires finite, consistently typed states/events and controllability; total deterministic monitors with absorbing errors; typed transfers and non-error initializers covering interval entry; and strict partial-order dependencies. Endpoints are safe, deadlock-free and uncontrollable-preserving, with controller alphabets covering environment labels; targets are nonempty and reachable. Justifying initializers against their declared requirement meaning is separate from game eligibility. Interval and post-handover duties belong to R^{upd} and R^{new} ; $\text{start}(r^{\text{new}}) < \text{stop}(r^{\text{old}})$ enforces overlap.

3.2 States, Transitions, and Completion

States and all entry states. An update state $q = (\xi, \mu, P)$ records each component's version/state $\xi(i) = (v, s)$ with $s \in S_{e_i^v}$, each monitor's state $\mu(t_r) \in M_r \cup \{\perp\}$ ($\perp$ means inactive), and pending commands $P \subseteq \Sigma_{\text{upd}}$. Here $\rho_i \in P$ exactly when component i is old. Old monitors remain active until their stop leaves P ; new monitors become active when their start leaves P ; update monitors are always active.

Admission $\text{Init}(z)$ copies z 's component and old-monitor states. It initializes interval monitors through $\iota_r(\xi)$, leaves new monitors inactive and sets $P = \Sigma_{\text{upd}}$. The policy replaces C^{old} , discarding its state and scheduling restrictions; undeclared objectives, including old liveness, are not preserved. Old monitors remain until stopped. The entry set is

$$Q_0 = \{\text{Init}(z) \mid z \in \text{Reach}(CL^{\text{old}})\}. \quad (2)$$

All-entry coverage avoids waiting for a favorable old state; an admission predicate would define a different contract. In the cell, $Q_0 = \{q_A^0, q_B^0\}$ distinguishes the two holders.

Four kinds of induced transitions. For $\xi(i) = (v, s)$, component i uses e_i^v . The active-component product determines enabled ordinary uncontrollables $U(q)$, including those with unsafe outcomes,

independently of updates and goals. For every $a \in \Sigma_{\text{upd}}$, impose

$$U(q) \neq \emptyset \implies \text{Post}(q, a) = \emptyset. \quad (3)$$

This global *uncontrollable quiescence* differs from transaction quiescence [18] and tranquility [37]. It is a command restriction, not a consequence of local transfer. Ordinary controllables may coexist with uncontrollables; the UC-priority lemma later justifies suppressing them. $\text{Post}(q, a)$ contains exactly the following outcomes; unmentioned coordinates remain unchanged. “Advance” means $\mu'(t_r) = \eta_r(\mu(t_r), a)$ for the specified active monitors.

Ordinary event $a \in \Sigma_N$. If a belongs to the local event set of at least one active component and every active component having a can transition, take every synchronous-product outcome and advance every active monitor.

Component update ρ_i . If $U(q) = \emptyset$, order-enabled, and $\xi(i) = (\text{old}, s)$, set $\xi'(i) = (\text{new}, s')$ for each $s' \in g_i(s)$, delete ρ_i from P , and advance every active monitor.

Requirement deactivation $\text{stop}(r)$. If $U(q) = \emptyset$ and order-enabled, advance every active monitor except t_r , retain inactive monitors at $\perp$, set $\mu'(t_r) = \perp$, and delete $\text{stop}(r)$ from P ; t_r does not observe it.

Requirement activation $\text{start}(r)$. If $U(q) = \emptyset$, order-enabled, and $\xi \in \text{dom } t_r$, advance every active monitor, retain other inactive monitors at $\perp$, install $\mu'(t_r) = t_r(\xi)$, and delete $\text{start}(r)$ from P ; t_r begins online observation afterward.

On the cell path starting with A holding, start preserves old/empty A and new/holding B , installs inspection-pending in the new monitor and removes itself from P . The old and interval monitors stay active.

If a precondition is false, the successor set is empty; every nondeterministic outcome is included and environment-chosen. Let $\mathcal{Q}$ contain every state reachable from Q_0 through these transitions. *Safety and handover-ready goals*. An update state $q = (\xi, \mu, P)$ is safe, written $\text{Safe}(q)$, exactly when every active monitor is outside its error states. For $P = \emptyset$ and $z \in Z_{\text{load}}$, write $q \sim z$ when component states (without version tags) and all new-monitor states agree. The handover map κ selects such a full endpoint state z . Goals are

$$Q_{\text{goal}} = \{q = (\xi, \mu, \emptyset) \in \mathcal{Q} \mid \text{Safe}(q) \wedge U(q) = \emptyset \wedge \exists z \in Z_{\text{load}} : q \sim z\}. \quad (4)$$

At a goal, all components and endpoint monitors are new; interval obligations end and matching new monitors continue. Handover disables ordinary controllables and installs $\kappa(q)$'s controller state, outside Σ and the rank. No uncontrollable can compete. A matching UC-enabled state is not a goal; compatible targets may differ in later safe behavior.

State-dependent policies and finite completion. A memoryless update policy is a function $\sigma : \mathcal{Q} \setminus Q_{\text{goal}} \rightarrow 2^{\Sigma_c}$ selecting only events with nonempty successor sets.

The possible next states under this policy are

$$\text{Succ}_{\sigma}(q) = \bigcup_{u \in U(q)} \text{Post}(q, u) \cup \bigcup_{a \in \sigma(q)} \text{Post}(q, a). \quad (5)$$

The environment chooses any uncontrollable and any permitted-event outcome. An uncontrollable self-loop blocks updates by Equation (3) and permits a losing infinite run. The returned policy retains one controllable at a quiescent state.

Let $\text{Reach}_{\sigma}(Q_0)$ be the policy closure from Q_0 , stopping at goals.

Runs, winning, and witnesses. A run follows the policy successors and stops at a goal. It is *maximal* if infinite or ending at a goal or a state without successors. A single policy σ is winning for an FG-DUCS instance if, from every $q_0 \in Q_0$, (i) every reachable update state is safe, (ii) every

reachable non-goal state has a successor, and (iii) every maximal update run reaches Q_{goal} after finitely many events. This *finite completion* counts observed transitions and assumes no fairness.

A *progress-ranking function* for a winning policy is $\text{rk} : \text{Reach}_\sigma(Q_0) \rightarrow \mathbb{N}$, with value 0 at goal states and satisfying $\text{rk}(q') < \text{rk}(q)$ for every non-goal state q and every $q' \in \text{Succ}_\sigma(q)$.

The FG-DUCS problem. The seven supplied groups define the eligible game. Decide whether one policy safely completes it from every entry under every permitted event and outcome. Return $(\sigma, \text{rk}, \kappa)$ covering all Q_0 , or a losing-region certificate.

3.3 Why the Generated State Represents the Contract

A *local update history* $\ell = z(a_1, o_1) \cdots (a_n, o_n)$ begins at an old closed-loop tuple and records each event and its realized local successors. Components follow their active-version transitions and local transfers; boundaries preserve physical states. Monitor memories fold their active intervals: stop excludes its own boundary; start installs its initializer after the boundary. Initially inherit old memories, initialize interval monitors and release old control, leaving new monitors inactive.

Ordinary events retain all combinations of participating local outcomes. Commands require UC quiescence, unexecuted status, completed predecessors and a transfer/initializer value where applicable. Quiescence includes unsafe UC outcomes. A source policy selects controllable labels and must retain every UC and every permitted outcome. Every maximal history must safely reach a quiescent matching target after all commands. Technical Appendix A gives the full recurrences.

PROPOSITION 3.1 (ADEQUACY OF THE GENERATED UPDATE GAME). *For an eligible contract, map each local update history to its current component tuple, active monitor memories and unexecuted commands. Histories with the same image admit the same next events and all the same local outcomes, agree on safety, and have the same handover targets. This map gives a label- and outcome-preserving correspondence between local update histories and paths of the generated game. A history-dependent policy satisfying the source obligations exists exactly when a winning policy exists in that game.*

PROOF. Fold a source history into its current tuple $F(h)$. At entry this is $\text{Init}(z)$. Each ordinary step uses the same participating component edges and full Cartesian product of outcomes; each transfer uses the same quiescence/order tests and g_i , advances active monitors and consumes its command. Stop removes its own monitor before observation; start installs its initializer after observation by existing monitors. Thus every label and outcome has exactly the same successor image, and safety and matching targets agree.

Induction maps source histories to game paths. Conversely, choose an old tuple witnessing the entry and lift each edge by this one-step correspondence, also for infinite paths. A game policy lifts through F ; applying a source policy to consistently lifted game histories gives a history-dependent game policy. Both directions preserve UC events, all selected-label outcomes, unsafe prefixes, deadlocks and infinite avoiding runs. Theorem 4.1 supplies a state-based winner or excludes every history-dependent winner, completing the equivalence. $\square$

This proves correspondence for the supplied contract; requirement meaning and Java frontend correctness remain separate obligations. Technical Appendix A gives the full source recurrences and expanded proof. The construction reconnects each goal to the supplied endpoint.

PROPOSITION 3.2 (PRESERVATION OF THE SELECTED ENDPOINT). *For a returned WIN certificate and reached goal q , the new-phase submodel reachable after handover at q is label-preserving isomorphic to CL^{new} rooted at $\kappa(q)$. Hence their finite and infinite ordinary-event traces coincide.*

PROOF. Handover retains matching components and monitors and installs the selected controller state. Each new-phase state copies an endpoint state with exactly its labeled transitions,

without further policy restriction or return to updating. This bijection preserves and reflects edges, hence finite and infinite paths. $\square$

Continuation properties established at the selected state are preserved; target selection can change which continuation applies. Execution additionally requires the adapter premises in Section 5.

Why matching and reconnection both matter. Cell’s prefix $\rho_B, m_B, \text{start}, \text{stop}, \rho_A$ finishes every command with inspection pending. Physical matching alone admits clear target z_B , whose m_A violates that duty; monitor matching excludes the reset. Conversely, disabling controllable loop b in a one-state endpoint with loops a, b and goal *true* preserves legality, deadlock freedom and safety [30, Defs. 3.9–3.10, 4.3], but loses b -continuations. Reconnection preserves them.

Technical Appendix O characterizes policy reuse under changed endpoints, with a saved PC2 specialization.

3.4 What the Local Interface Generates

Table 2 distinguishes the supplied structures from the generated result. DUCS already constructs an update environment and goal; GR(1) updating already constructs a bridge. From the supplied local contract, FG-DUCS constructs mixed-component transitions, implements individual monitor lifetimes, and derives handover target tests.

Table 2. Input and generation responsibilities. Existing guarantees are retained in the comparison; the rows do not claim a general translation or measured modeling-effort reduction.

Interface	Supplied structure	Generated result and guarantee
DUCS [30, Defs. 4.1–4.3, 5.1]	Environments/goals, old controller, global transfer, transition requirements and update fluents	Update environment/goal, update-and-subsequent controller and controller-transfer relation; legal, deadlock-free completion from every old-controller state
GR(1) update [2, Def. 3.1, Secs. 4–4.1]	Old/new specifications and switching condition	New controller, bridge and switching book-keeping; bounded switching and new-specification satisfaction from winning states
FG-DUCS	Local components/transfers, requirement lifetimes/initializers and fixed endpoints	Mixed game and WIN/LOSS certificates; local-history correspondence and selected-endpoint continuation

Concrete Cell constructions. Our DUCS witness supplies eight mixed stages, three endpoint states, empty-station guards, inspection/lifetime fluents and a target gate; DUCS generates its protocol. The GR(1) witness supplies version/lifetime variables, $p' = [h = B]$ at start, release at stop and a target test; its algorithm generates switching bookkeeping. Both admit the Cell paths; the appendices distinguish their endpoint guarantees. Technical Appendices K and P give the full analytical constructions, including the DUCS eventual-handover condition beyond native safety- T [30, Sec. 5.3]. These are neither solver runs nor general translations or encoding-size lower bounds.

4 Synthesizing and Checking an Update Policy

Lazy explores products on demand using finite reachability reasoning [7, 8, 36].

A *bucket* is $\text{Post}(q, a)$; an empty bucket disables a . *UC priority*: disabling controls at a UC-enabled state preserves winning; it removes runs and creates no deadlock because every UC outcome remains (Lemma A.1). *Memoryless sufficiency*: whenever a history-dependent policy wins, a state-based one wins; the proof below establishes this by attractor ranks and a history-independent losing-region response.

4.1 Lazy Successor Generation

Algorithm 1 constructs a reachability attractor [5]. It tracks discovered states V , states $X \subseteq V$ with all UC candidates queried, and proved winners W . States in X may retain unqueried controls; W may omit winners.

Expansion queries every UC candidate. If none is enabled, it queries controls through the next nonempty bucket or exhaustion; Retry resumes unfinished cursors. Each query retains all outcomes. Candidates include local events and precedence-enabled commands in any fixed finite order.

Promote. A safe expanded state wins when its nonempty proving set—all UC successors, or one control bucket at quiescence—lies in W ; its rank is one above their maximum. Reverse incidences trigger predecessor reconsideration. Failure requires exhausted candidates and a promotion fixed point. Once all entries win, extraction selects one lower-rank control bucket at each reached quiescent non-goal, none elsewhere; it need not follow promotion or minimize length.

Algorithm 1: Lazy

Input: Typed update contract (Section 3)

Output: $(\sigma, \text{rk}, \kappa)$ or (no solution, L)

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1   $V \leftarrow Q_0$ ;  $X, \text{Retry} \leftarrow \emptyset$ ;  $W \leftarrow V \cap Q_{\text{goal}}$ ;
2   $\text{Open} \leftarrow \{q \in Q_0 \mid \text{Safe}(q) \wedge q \notin Q_{\text{goal}}\}$ ; set  $\text{rk}(q) \leftarrow 0$  for  $q \in W$ ;
3  while true do
4      Add queued goals to  $W$  once with rank zero; exhaust Promote, retaining witnesses/ranks, removing winners
        from Retry and propagating predecessors;
5      if  $Q_0 \subseteq W$  then return  $(\sigma, \text{rk}|_{\text{Reach}_{\sigma}(Q_0)}, \kappa)$  constructed from the goal witnesses and retained lower-rank
        choices;
6      if  $\text{Open} \neq \emptyset$  then
7          Remove  $q$  from Open, add it to  $X$ , and query all uncontrollable candidates;
8          Close the control cursor if UC-enabled; otherwise initialize it and query to the next nonempty bucket or
            exhaustion;
9          Insert  $q$  into Retry iff  $q \notin W$  and its cursor has remaining candidates;
10     else
11         if  $\text{Retry} = \emptyset$  then return (no solution,  $L \leftarrow V \setminus W$ );
12         Remove  $q$  from Retry; query to the next nonempty bucket or exhaustion; reinsert iff  $q \notin W$  and its
            cursor has remaining candidates;
13     Add every queried successor to  $V$ ; queue newly discovered goals for rank zero and safe non-goals for Open
        (unsafe states stay outside  $W$  and Open);
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4.2 Correctness and Certificates

THEOREM 4.1 (TERMINATION, SOUNDNESS, AND COMPLETENESS). *For every eligible finite FG-DUCS instance with Equation (3) and quiescent goals in Equation (4), Algorithm 1 terminates. A returned (σ, rk, κ) keeps every active monitor non-error, preserves every uncontrollable event and every outcome of each policy event, reaches Q_{goal} finitely without deadlock, and gives each reached goal a compatible target in Z_{load} for the supplied game. It returns no solution iff no winning policy covers Q_0 in that supplied game.*

PROOF. *Coverage.* Initialization/discovery queues every new safe non-goal in Open. Only expansion removes it, after querying all UC candidates. At quiescence, expansion/retries advance a finite control cursor and retain every nonwinner with candidates left in Retry; promotion removes only winners. Thus all expanded states have complete UC outcomes, unfinished quiescent nonwinners remain retryable, and unexpanded safe non-goals remain open.

Termination. Each state is expanded/promoted once and each candidate queried once. Reverse incidences are processed on promotion, so the finite search terminates.

Soundness. Goals start at rank zero with compatible handover states. By induction, each promotion adds a safe state above a complete nonempty proving set of previously ranked winners: every UC successor, or every outcome of one control event. Queried buckets are immutable. Extraction retains all UC events, chooses one such control bucket at quiescence, and disables other controls. Every reached non-goal is safe, has a successor, and every successor has strictly smaller natural-number rank. Consequently no maximal run is an infinite run or a non-goal deadlock, and every run reaches a matching quiescent goal. The monitor rules advance exactly the memories active over each event; induction from entry establishes every declared safety interval. Handover retains the matching new memories.

Completeness. At failure both exploration queues are empty and promotion has reached a fixed point. Some entry is in $L = V \setminus W$, and L contains no goal. Coverage ensures that each safe state in L was expanded. If it is UC-enabled, at least one UC outcome lies in L , since otherwise it would be promoted. At quiescence every control candidate is exhausted; each enabled control bucket has an outcome in L for the same reason. Against any history-dependent policy the environment can select such an outcome after every history. Disabling all events deadlocks; otherwise the resulting run reaches an unsafe state or stays outside the goals forever. Hence no winning policy covers that entry. A winning policy therefore rules out failure and, by termination, forces success. Finally, UC priority loses no winning policy: removing optional controls at a UC-enabled state removes runs without introducing a deadlock, and the restricted policy remains admissible in the original game. $\square$

Checking the returned evidence. WIN checking requires exactly $D_\sigma = \text{Reach}_\sigma(Q_0)$ as the rank domain, safety, enabled selected controls, nonempty all-outcome closure and strict decrease at non-goals. Goals require rank zero, quiescence and a compatible κ , defined exactly on $D_\sigma \cap Q_{\text{goal}}$; omitted policy entries are empty. LOSS checking requires an entry-containing, goal-free L with the UC/all-control counter-outcomes used in the completeness proof. These certify search relative to the successor and endpoint predicates [17]; Section 6 states their implementation trust boundary.

4.3 Exploring the Compiled Contract on Demand

With indexed predicates and constant-time bookkeeping, search is linear in discovered states, queried candidates and returned outcomes, excluding preprocessing and Java sorting/scans. Lazy may explore the whole reachable graph. For independent one-state updates, transfer-first Lazy

instead visits $n + 1$ states versus Direct-Full’s 2^n subsets; Technical Appendices E and J give the accounting and complete default-order proof.

5 Interpreting and Executing a Winning Policy

The construction proves properties of the supplied game. Requirement meaning and execution conformance are separate obligations; Table 3 states the evidence for each before the evaluation.

Table 3. Three guarantees and the evidence needed to apply them. Technical Appendix M.1 and S3–S4 give the individual requirement checks.

Layer	Established result	Application obligation
Game	Every policy run safely completes from every entry and matches a supplied target	Theorem 4.1; certificate checks use the supplied successor and endpoint predicates
Requirement	Active monitors enforce their declared continuation languages	Cell, PC2 and GSM have direct duty arguments. Saved lookup/source and entry checks support other initializers; actual key/observation conformance and intended scopes remain premises
Execution	Observed execution follows the policy and finishes within its entry rank	A1–A4 below connect observations and atomic commands to the model; execution conformance and real-time bounds are unvalidated

5.1 Giving Requirements Their Intended Lifetimes

An initializer assigns the obligation present at activation; Cell illustrates why this is more than selecting a safe state.

An exact activation obligation. Cell’s owed-inspection bit starts at 1 exactly when B holds the workpiece. Event m_B sets it, j clears it, and m_A is forbidden while it is 1; other labels preserve it. Encoding 0 by c and 1 by d makes every monitor transition identical to this rule, including rejection of forbidden departure. Initialization and suffix induction prove exact enforcement. Figure 1’s first path starts with debt and clears it by ordinary inspection; the second starts clear.

Obligations that begin at activation. ACT starts $\Box\varphi$ at the actual post-start observer vector. A faithful invariant monitor reached safely with that vector has exactly the ACT continuation language (Proposition M.2); the reference word need not be plant-executable. Source checks prove this correspondence for 900 non-error cells in 138 saved tables. Application requires the actual key, matching future observations and no fallback. PC2 and GSM have direct policy arguments below.

When prior histories are part of the requirement. For prefix-closed requirement language S_r , declared activation histories $H_r(\xi)$ and monitor continuations $L_r(m)$, residual soundness (RS) means $L_r(l_r(\xi)) \subseteq \bigcap_{h \in H_r(\xi)} \{w : hw \in S_r\}$. New histories include start once; monitors consume later events. Equality is exact; strict inclusion can reject feasible updates, and an empty history set makes inclusion vacuous. Scope completeness and intent are external obligations (Technical Appendix B.1).

A *fluent* records setting/resetting events. The saved checks use **A** for NEW (safe pre-activation histories consistent with the component observation) and **E** for UPD (reference beginning at entry).

H, which counts prior violations too, is not claimed (S3). Cell’s pending initializer is exact for occupancy-based debt but conservative for a start-only reference that starts clear. A inclusion alone establishes neither the occupancy duty nor ACT.

5.2 Execution and Handover

The execution connection has four distinct premises (Technical Appendix H):

A1: endpoint observations preserve component/monitor states and same-labeled endpoint edges.

A2: update control disables only unselected controllables, preserves every UC event/outcome, reports the successor and eventually observes a next event.

A3: commands start quiescently and finish atomically, finitely and failure-free with an observed modeled successor.

A4: entry observes/copies the actual old state; quiescent handover installs the selected compatible state, using finite failure-free linearizable boundaries [15].

Under A1–A4, every execution maps to the linked model and reaches handover within the entry rank (Theorem H.1). History completeness means that every actual activation history, with its declared reference origin, belongs to the supplied history scope; with RS it justifies the residual interpretation. No event-choice fairness is assumed. Internal divergence and unobservable outcomes are excluded; rank bounds events, not time. Table 3 separates these premises from the scope checks and proofs actually obtained.

6 Implementation and Evaluation

6.1 Design and Implementation

Four FG procedures use 27 contracts from 9 inputs. Base has no interval requirement; R1 restricts ordinary actions and R2 orders boundaries versus transfers. PC2-Rolling adds continuous availability. Legacy, our DUCS/GR fork, synthesizes subsequent control instead.

Implementation. Our Modal Transition System Analyser (MTSA) extension [9] checks certificates [26]; Link checks old–update–new composition, entries and targets. An independent oracle parses restricted Finite State Processes (FSP). Finite-JSON runs save losing membership; FSP benchmark runs report only its statistics. Runtime conformance is unchecked.

Use of generative AI. Generative AI assisted contract and example design, proof drafting, Java solver/frontend implementation, Python checkers, experiment drivers, execution, result analysis and manuscript editing. Validation used certificate and exhaustive checks for algorithms, source and automaton comparisons for initialization, saved-output checks for tables and primary sources for citations. Claude Opus 5.5 and ChatGPT Pro/Latest supplied editorial reviews, not empirical evidence. Authors retain responsibility; shared AI/code can produce correlated faults. Technical Appendix N details tool roles; no human evaluation was performed.

Constructing the benchmark inputs. Nine inputs adapt eight DUCS sources; PC1/PC2 share a parent. Adaptations make PC tool results UC, shorten Railcab transfers preserving both outcomes, omit PowerPlant’s initial ENV pair, and change safety/ordering; equivalence is unestablished. Technical Appendix J specifies every adaptation.

FSP goals and component/transfer declarations feed MTSA’s GR endpoint backend. Targets are all reachable new-endpoint states. The frontend compiles monitors, enumerates physical closure and fills global-tuple initializer tables from current observers; interval tables use post-entry values. Observers track nonlocal labels and survive commands/transfers. Lazy generates the remaining product on demand; JVM cost includes preprocessing.

Inputs have 1–5 components, 4–131 entries and 3–26 commands (Technical Appendix C). Only Railcab transfers branch; none declares precedence. R2 encodes order in monitors. GSM/PC1’s single transfer makes merging the identity.

Exploration variants. All methods share entries, successors, goals, checks and terminals. The on-the-fly (OTF) variants omit controls at UC-enabled states and stop when all entries win. Direct-Full first materializes reachable buckets including those controls, then solves with the same UC-priority rule. Full+UC is an auxiliary full-construction variant that omits those controls during construction. At quiescence Lazy queries individual buckets, Eager all controls, and Update-first favors updates, testing O-DUCS’s priority [16] within our solver. Lazy favors stops/transfers, prioritizes states by endpoint mismatch and pending-command counts, and uses LIFO exploration before FIFO retries (Technical Appendix E); this order need not minimize exploration.

Design and measures. RQ3 uses one Windows/Xeon host, 64 GiB heap and 1,200 s JVM cap: 27×4 same-game cells plus 27 Legacy references. Medians require five completions; censored cells retain their capped trial. Solver time includes checking; total JVM adds frontend, startup, output and endpoints. We also report exploration, policy size and peak working set. Legacy uses its native timer; Mac pilots are excluded. Full settings and interruptions remain in Technical Appendices N and J.8 and S4.

6.2 RQ1: Can Local Choices Change Feasibility?

Cell and Policy isolate component and requirement-lifetime obstructions (Section 2). These new cases exercise choices absent from the inherited DUCS-oriented benchmarks. We test whether merging those choices, with the remaining contract fixed, removes all completing policies; this is a mechanism question, not a frequency estimate over deployments.

What is merged. Partition transfers, old stops and new starts separately into blocks. A transfer block $B \subseteq I$ simultaneously permits all targets in $\prod_{i \in B} g_i(s_i)$; a boundary block removes or initializes its monitors at the same component tuple. Other active monitors compose member-label effects, checked to commute pairwise on every monitor/residual state. Mapped precedence must remain irreflexive and acyclic. Ordinary behavior, entries, endpoints, relations, initializers and targets stay fixed; quiescence and all member guards apply. These are comparison contracts, not DUCS encodings.

The following obstruction explains the Cell, Policy and PC2 merges directly; the rolling threshold then gives an exact quantitative boundary.

LEMMA 6.1 (MANDATORY-COMMAND OBSTRUCTION). *Let q_0 be an entry and B a nonempty set of commands pending there. Suppose a safe region R contains q_0 , retains all commands in B , and is closed under every safe successor whose label is outside B . If each command in B is disabled or has an unsafe outcome at every state in R , no winning policy covers q_0 .*

PROOF. Every goal exhausts pending commands, so a winner must execute a first command in B . Safety and closure keep its prefix in R , where that step is disabled or has an environment-chosen unsafe outcome. Avoiding it yields a non-goal deadlock or infinite run, also losing. $\square$

In Cell’s one-workpiece region, joint transfer is disabled. Policy retains its initial active bits until either grouped boundary violates the interval. PC2-Rolling’s enabled joint transfer violates availability (Section 6.2.1). These results concern specified merges, not native DUCS expressiveness.

Mechanisms and scope. Readiness/rejoining gates motivate staged replacement [28, 34]. Rolling begins with n ready replicas and no spares; its interval requirement keeps at least m ready. Each

transfer boots its replicas; each has one UC readiness event and no waiting loop. Service self-loops are controllable. The target is all-new and ready.

PROPOSITION 6.2 (ROLLING THRESHOLD). *For $n \geq 2$, $1 \leq m < n$ and $1 \leq k \leq n$, fixed consecutive batches of size k (with a smaller final remainder) admit safe finite completion iff $k \leq n - m$.*

PROOF. Some mandatory batch has size k . Its transfer leaves at most $n - k$ ready replicas, so $k > n - m$ violates the interval. Otherwise, transfer one batch only when all replicas are ready, disable service, and retain every UC readiness outcome before the next transfer. Readiness never falls below $n - k \geq m$. With u old and b booting replicas, $2u + b$ strictly decreases on every selected transfer and every UC event, reaching zero exactly at the target. Thus all maximal runs finish without fairness. $\square$

The 15 Rolling separations compare $k = 1$ with $k = n$; the 70 threshold cells are separate. Canary adds health reports/recovery, DB-Rolling role-sensitive replacement, and Rolling+Audit overlapping duties. Policy isolates opposing boundaries; Threads supplies both-WIN/both-LOSS quiescence controls. Of the 33 separations, 30 are parameter points in Rolling and Canary; they are not 33 independent mechanisms. Table 4 keeps all outcomes. Technical Appendix J.11 specifies mechanisms and assumption controls; their deployed prevalence is unestablished.

Answer. The suite yields 33/55 fine-WIN/merged-LOSS comparisons across five families. PC2 contributes no solver WIN/LOSS pair because its merged run times out; the source argument below independently proves merged impossibility.

Why finer is not always better. Two components with one state per version share UC labels x, y . The only UC loops are y in old 1, x in new 1, x in old 2 and y in new 2; shared labels synchronize. Thus all-old and all-new are quiescent, but either mixed product has a UC loop. With identity transfers and no requirements, either first local transfer prevents quiescence forever, so fine loses. A joint transfer skips the mixed state and wins at rank one. A shared controllable loop makes both endpoints deadlock-free. Atomicity can therefore remove either a necessary or a harmful intermediate state; there is no general refinement order (Technical Appendix J.12).

Table 4. Constructed granularity comparisons: each pair fixes one model and compares its fine and merged contracts. Policy merges boundaries; other rows merge transfers. n counts replicas/workers/stations. Canary bounds reported failures by $n - m$; otherwise m is an availability floor. The main total excludes Cell.

Family / control	Parameter domain	Pairs	WIN / LOSS	LOSS / LOSS	WIN / WIN	Inc.
Rolling	$n = 2 \dots 6$, $1 \leq m < n$	15	15	0	0	0
Canary	$n = 2 \dots 6$, $1 \leq m < n$	15	15	0	0	0
Policy	2 rule pairs	1	1	0	0	0
DB-Rolling	$n = 3$, $m = 2, 3$	2	1	1	0	0
Rolling+Audit	$n = 2$, $m = 1$; 1 audit pair	1	1	0	0	0
Threads (control)	$n = 2 \dots 6$; queue 1,2	20	0	10	10	0
PC2-Rolling	2 arms; floor 1	1	0	0	0	1
Main total		55	33	11	10	1
Cell _n (reference)	$n = 2 \dots 10$	9	9	0	0	0

Completed columns read fine / merged; Inc. denotes unresolved. PC2's merged solver remains TO, not LOSS. Threads is a negative control; its queue capacities are 1 and 2, with both backpressure and saturated-offer regimes. Rolling+Audit counts transfer merging; boundary merging remains WIN and adds no pair here.

6.2.1 *A Source-Derived Calibration Case.* PC2-Rolling adds calibration before replacement arms resume production, requiring another arm operational. Calibration requests are controllable; completions are UC; transfers enter calibration states. Erasing calibration recovers the original operational process, without whole-problem equivalence. Fixed endpoints retain baseline requirements plus availability, giving 13 new starts. Endpoint availability tracks requests/completions; the interval also treats transfers as calibration entry. Transfers preserve initializer observers. The 25 boundaries and two transfers have no supplied order; Technical Appendix L compares generation responsibilities.

Generated update and certificate. The saved policy calibrates arm 1, waits for completion, then calibrates arm 2 and waits again. Transfers directly enter calibration, so no calibration request occurs. All 126 entries are covered; every outcome decreases rank from at most 37 and keeps 1 arm outside calibration. Link checks the full old–update–new composition. Table A7 and complete counts are in Technical Appendix L; this replay supplies no timing.

Source obligations through handover. Arm 1 retains $\Box(out_1 \Rightarrow Drilled_1 \wedge Cleaned_1 \wedge Painted_1)$. Successful tool results set its initially false fluents; reset clears them. At the eighth start on the saved path, no tool has succeeded. Initialization installs zero observers, so departure still requires all three tasks; falsely marking them finished would allow violating out_1 .

Every run from all 126 entries traverses the same thirteen-start chain. Each source invariant holds at its start and is preserved by the remaining starts until the endpoint takes over, proving all thirteen ACT obligations on this policy’s reached tuples. No ordinary event occurs on that suffix; GSM separately shows active ordinary behavior. This does not cover all initializer tuples or all 138 occurrences (Technical Appendix L.1).

Only 39/126 entries admit a safe full prefix; 87 fail (21 before entry, 66 during update). All admit an earlier violating cycle restoring zero observers. These limit history-wide interpretations, not ACT. Safe-history nonemptiness for 38,288 initializer tuples and runtime remain unchecked (Technical Appendix L).

PC2-Rolling: fixed-merge obstruction. All checked old entries admit both raw transfers. Before their mandatory joint transfer, old plants lack calibration actions; ordinary moves and boundaries preserve readiness even after releasing old control. Every joint outcome calibrates both arms and clears both interval readiness bits, reaching absorbing error. This source argument proves impossibility independently of solver completion. The merged solver and fine Direct-Full nevertheless time out at 7200 s/200 GiB; these are not losing decisions.

Inherited inputs and additional witnesses. On nine inherited inputs, merging transfers, boundaries or both changes 0/54 Base/R1 decisions. Their total raw transfers omit Rolling’s startup phase; totality alone does not ensure safe merging. States decrease in 44 comparisons and stay equal in 10. R2 names individual events and is excluded. This null result limits observed granularity benefits.

Additional restricted-transfer comparisons retain both-LOSS and incomplete cases separately (Technical Appendix J; S1–S4).

6.3 RQ2: Do Policies Satisfy the Contract?

Winning and losing applications. Railcab has four component pairs, twelve requirement boundaries and a transfer with ENDOFTS/AC outcomes. Two new-AC states differ in approachingCrossing memory, so location alone determines neither initialization nor target compatibility. All 22 entries of each saved policy stop every old requirement before the first new start: Base allows 1–4 ordinary events in this unprotected interval, whereas R1 allows none and retains seven interval guards. Both pass certificate/Link checks. Independent lifetimes thus require the modeler to state

any continuously needed obligation as an interval requirement; WIN alone does not supply it. These exports do not characterize all benchmark gaps (Technical Appendix J.9; S4).

Industry/R1 forbids validation between old-stop and new-start and loses at 42/131 entries by exhaustive checking. Its trap offers violating GF1/TOR validation or an infinite DSD1 rejection cycle; disabling DSD1 deadlocks. The losing closure excludes all orders. Native DUCS and its aligned fork win on their different subsequent-control problem [30, Defs. 4.1–4.3]. Exempting GF1 yields 131 WIN through earlier paths, not trap escape (Technical Appendix J); these checks supply no timing.

Answer. On the finite validation fixtures, Lazy and Direct-Full match all expected results from independently reconstructed labeled transition systems (LTSs). Application certificates additionally check the shared Java game semantics; source-level initialization evidence is reported separately below.

Independent reconstruction and certificates. Python independently constructs graphs and certificates for 46 fixtures: $46 \times 2 = 92$ jobs yield 36 WIN, 48 LOSS and 8 invalid (missing events, deadlocks or suppressed UC). All 84 eligible jobs pass certificate/exhaustive checks; all WIN jobs pass Link. Explicit successors are separate; frontend and endpoint/goal code are shared. These jobs are not independent applications.

Six monitor-free fixtures independently reconstruct restricted FSP, endpoints, complete games and goals. All 138 NEW tables match 1,587 reconstructed entries. A separate source parser checks the 46 invariant positions across three variants: initial truth, source observer rules and all 35,283 transitions from 900 safe monitor–observer pairs agree. Induction gives source-invariant fidelity for these saved objects and their checked alphabets, not arbitrary FSP compilation, physical decoding or actual activation conformance (Technical Appendices G and M.1).

Checking initializer interpretation. There are 138 safe-history NEW inclusions and 135 entry-scoped UPD justifications (46 R1/89 R2), distinct from ACT. Empty scopes pass vacuously. Technical Appendix G gives all formula, label, reset and automaton checks.

Source obligation over ordinary actions. GSM changes $\Box \neg \text{decode.3}$ to $\Box \neg \text{decode.2}$. From every old entry, reach the first RECEIVED state; then

$$\text{RECEIVED} \xrightarrow{\text{stop,start}} \text{RECEIVED} \xrightarrow{\text{decode.3}} \text{DECODED} \xrightarrow{\text{out put}} \text{SENDER} \xrightarrow{\rho} \text{SENDER}_{\text{new}}.$$

The preceding *receive* sets the *decode.2* observer to zero; boundaries preserve it. The saved initializer maps zero to monitor state zero: *decode.2* enters error, all other labels preserve zero. Thus the new obligation constrains an actual ordinary choice. Transfer matches sender, observer and new-monitor zero to the endpoint. The pre-stop distance is at most seven; adding the five displayed events gives rank at most twelve. Consecutive stop/start satisfies R1’s gap ban. This source-derived analytical policy proves ACT, not a new solver result; a separate 45-pair source check establishes E initialization (Technical Appendix M).

The 25 WIN certificates have 8–1,071 update states and maximum ranks 4–34, consistent across five repeats. Linked old–update–new controllers have 15–1,260 states (S4). Ranks bound events, not downtime or optimal length (Technical Appendix J.8).

6.4 RQ3: Does Lazy Reduce Exploration?

Answer. Lazy avoids states and queries in all 14 completed Direct-Full pairs. Within the fixed budget it decides 26 contracts, Eager 17 and Direct-Full 14; all paired decisions agree (Table 5). Update-first is faster on 8/24 paired solver medians but adds two timeouts. Lazy returns LOSS for

Industry/R1 and times out on PC2/R2 at 1,200 s/64 GiB (fixed), 3,600 s/64 GiB (sensitivity) and 7,200 s/200 GiB (S4.6, earlier series); the limiting resource is unknown.

Legacy yields 9 WIN, 13 OOM and 5 unmeasured snapshot-cap cases. Its differing subsequent-control objective precludes same-game ratios or impossibility inferences (Technical Appendix J).

Table 5. All 27 contracts per method. TO denotes one capped trial, not a timing median. Ratios summarize heterogeneous completed pairs; values above one favor Lazy.

Method	WIN/LOSS/TO	Pairs	Faster than Lazy	Solver ratio	Total JVM ratio
Lazy	25/1/1	—	—	—	—
Eager	17/0/10	17	0	2.04 [1.18–4,575.05]	1.60 [1.00–144.85]
Update-first	23/1/3	24	8	1.07 [0.19–39.76]	1.00 [0.32–6.10]
Direct-Full	14/0/13 [†]	14	0	9.14 [1.58–2,123.81]	2.27 [1.00–130.66]

TO: timeout; no OOM/unmeasured cases. [†] Workflow/R2 retains its interruption and later same-setting timeout (S4).

Comparator/Lazy ratios are median [min–max] across completed pairs of five-run medians. Faster counts lower comparator solver time. JVM includes frontend, startup, endpoints and output; solver includes checking. Unresolved costs have no ratio.

Figure 2 plots all 14 completed pairs; Appendix D gives exact states, queries and costs. In R2, Update-first uses fewer states and a lower solver median on six of eight completed pairs, showing order sensitivity (Technical Appendix J).

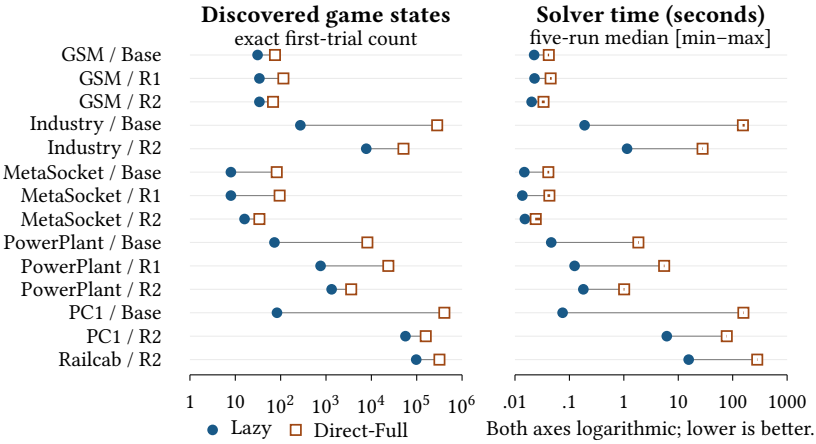


Fig. 2. All 14 completed WIN/WIN pairs. Solver time includes checking but excludes frontend, endpoint preparation and JVM startup. The other 13 Direct-Full contracts time out (Figure A4) and are omitted. This compares whole procedures.

Capped Direct-Full attempts give total-JVM lower bounds, not solver ratios or timing medians (Technical Appendix J.10).

Separating construction choices. Lazy versus Eager isolates deferred controllable-bucket generation under the same UC pruning, input, host and budget: 26 versus 17 decisions. Direct-Full additionally changes pruning. Auxiliary Full+UC timings cannot isolate that effect across differing builds, hosts and repetitions; no unpruned OTF is implemented (Technical Appendix J.10).

Scaling limits. All methods finish the independent family for $n = 2, \dots, 20$. At $n = 20$, Lazy and Update-first discover 21 states each, Eager 211, and Direct-Full 1,048,576 (S4).

Travel reaches timeouts and out-of-memory failures under all methods, including expanded-budget timeouts. These separate trials never enter medians; Technical Appendix J.10 retains all populations and adverse outcomes.

7 Related Work

Zhang and Cheng formalize temporal adaptation semantics [39]; scenario-based synthesis uses history-based updatability [12, 24]. Local updating [23], PROTEOS [13, Secs. 4.3–4.6], workflow updating [33, Secs. 4.3, 5.2], Kitsune [14, Sec. 2] and network synthesis [25] address replacement, migration or transition safety. Live Synthesis carries unfinished obligations [11]; our initialization reasoning uses scopes [10] and residual rewriting [32]. State attraction bounds startup steps under weak invariance [31]; compositional/modular control addresses specification structure [27, 38]. OTF-DCS requires marked extensions [8, Sec. 3], whereas FG requires every maximal update run to reach handover.

Ma et al. preserve transaction version consistency using dynamic dependencies and a sufficient freeness condition, with concurrent versions and selective blocking [21, Secs. 4.1–4.3]. Their transaction quiescence differs from FG’s UC quiescence; FG’s finite handover contract does not establish transaction consistency.

O-DUCS separates update and post-update analysis [16, Sec. 3]. Our fork’s plain OTF route precomputes a new controller and reconnects matching environment/safety states [4]; identity with the 2024 release is unestablished. FG combines this existing reconnection with supplied endpoints/targets, local transfers, individual lifetimes and initializers.

Thuijsman and Reniers [35, Secs. 4–7] support individual/simultaneous reconfiguration, feature-dependent requirements and state retention/reset. Their nonblocking supervisor permits a marked continuation from every reachable state. FG instead requires every maximal update run to finish at a supplied controller, with independent lifetimes and declared initialization.

8 Threats to Validity

Permanently enabled UC arrivals or polling block even local commands. Threads contrasts finite UC phases with perpetual activity. Fairness-dependent waiting is outside the contract; inherited-adaptation effects on feasibility are not isolated. Decoding, history completeness and A1–A4 remain unvalidated; repair is excluded. Constructed cases assume finite startup, reporting, recovery and election; selection may favor separations, and deployed prevalence is unestablished.

Shared authorship, AI and code risk correlated faults; independent parsing covers restricted FSP. Direct-Full is in-house; Legacy’s objective differs. Pruning isolation, external solvers and randomized orders remain untested. Ratios omit unresolved costs; extended trials lack medians. Sampling can miss peaks; RSS identifies neither heap occupancy nor bottlenecks. S4 retains interruptions.

Null/LOSS results, faster Update-first, lower Eager/Update-first memory and Travel/PC limits bound gains. PC2’s merged timeout remains unresolved. Fixtures/repeats are not independent applications; timing uses one machine. Inherited inputs lack precedence; only Railcab transfers branch. Exponential separation assumes the default order.

Conclusion. FG-DUCS compiles local transfers and requirement lifetimes into safe, completing policies preserving continuations from the selected new state.

Data Availability. The anonymous evidence snapshot [4] provides source, 27 application contracts, scripts and saved results. The submission contains this main paper and an integrated supplement (Technical Appendix A–P and Supplement S1–S5), with complete proofs, model definitions, result tables, bookmarks and continuous pagination. The repository’s claim map identifies checks and build commands; saved-evidence checks run without new performance experiments. Dependencies come from official upstream sources.

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